Lab: Columnar Example in MongoDB

1. Create Database

```
>_MONGOSH
> use("student_lab_db");
< switched to db student_lab_db
```

2. Create Collection and Insert Data

```
> db.student_results.insertMany([
    {
      student_id: 1,
      name: "Aung Aung",
      class: "CS",
      gender: "Male",
      subjects: {
        database: 85,
        programming: 90,
        networking: 78
      },
      semester: "2026-S1"
    },
    {
      student_id: 2,
      name: "Su Su",
      class: "CS",
      gender: "Female",
      subjects: {
        database: 92,
        programming: 88,
        networking: 81
      },
      semester: "2026-S1"
    },
    {
      student_id: 3,
      name: "Mg Mg",
      class: "IT",
      gender: "Male",
      subjects: {
        database: 70,
        programming: 75,
        networking: 80
      },
      semester: "2026-S1"
    }
]);
< {
    acknowledged: true,
    insertedIds: {
      '0': ObjectId('6ace0f17bb582898c653450c'),
      '1': ObjectId('6ace0f17bb582898c653450d'),
      '2': ObjectId('6ace0f17bb582898c653450e')
    }
  }
```

3. View Data

```
>_MONGOSH
> db.student_results.find();
< {
    _id: ObjectId('6ace0f17bb582898c653450c'),
    student_id: 1,
    name: 'Aung Aung',
    class: 'CS',
    gender: 'Male',
    subjects: {
      database: 85,
      programming: 90,
      networking: 78
    },
    semester: '2026-S1'
  }
  {
    _id: ObjectId('6ace0f17bb582898c653450d'),
    student_id: 2,
    name: 'Su Su',
    class: 'CS',
    gender: 'Female',
    subjects: {
      database: 92,
      programming: 88,
      networking: 81
    },
    semester: '2026-S1'
  }
  {
    _id: ObjectId('6ace0f17bb582898c653450e'),
    student_id: 3,
    name: 'Mg Mg',
    class: 'IT',
    gender: 'Male',
    subjects: {
      database: 70,
      programming: 75,
      networking: 80
    },
    semester: '2026-S1'
  }
```

4. Create Columnstore Index

```
> db.student_results.createIndex(
    { "$**": 1 },
    { name: "student_columnstore_index" }
  );
< student_columnstore_index
```

5. Check Index

```
> db.student_results.getIndexes();
< [
    { v: 2, key: { _id: 1 }, name: '_id_' },
    { v: 2, key: { '$**': 1 }, name: 'student_columnstore_index' }
  ]
```

6. Query Only Required Columns

```
> db.student_results.find(
    { class: "CS" },
    {
      _id: 0,
      name: 1,
      class: 1,
      "subjects.database": 1
    }
  );
< {
    name: 'Aung Aung',
    class: 'CS',
    subjects: {
      database: 85
    }
  }
  {
    name: 'Su Su',
    class: 'CS',
    subjects: {
      database: 92
    }
  }
```

7. Aggregation Example

```
> db.student_results.aggregate([
    {
      $match: {
        class: "CS"
      }
    },
    {
      $project: {
        _id: 0,
        name: 1,
        database_mark: "$subjects.database",
        programming_mark: "$subjects.programming"
      }
    }
  ]);
< {
    name: 'Aung Aung',
    database_mark: 85,
    programming_mark: 90
  }
  {
    name: 'Su Su',
    database_mark: 92,
    programming_mark: 88
  }
```

8. Explain Query Performance

```
> db.student_results.find(
    { class: "CS" },
    {
      _id: 0,
      name: 1,
      class: 1,
      "subjects.database": 1
    }
  ).explain("executionStats");
< {
    explainVersion: '1',
    queryPlanner: {
      namespace: 'student_lab_db.student_results',
      parsedQuery: {
        class: {
          '$eq': 'CS'
        }
      },
      indexFilterSet: false,
      queryHash: '9839872B',
      planCacheShapeHash: '9839872B',
      planCacheKey: 'BE666D93',
      optimizationTimeMillis: 0,
      maxIndexedOrSolutionsReached: false,
      maxIndexedAndSolutionsReached: false,
      maxScansToExplodeReached: false,
      prunedSimilarIndexes: false,
      winningPlan: {
        isCached: false,
```

===explanation ==

MongoDB command executed successfully.

Memory limit for many aggregation stages = 100 MB.

Query engine mode = try SBE when possible.

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